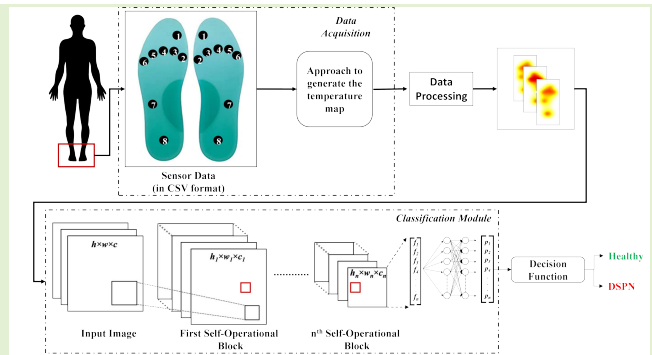


DSPNet: A Self-ONN Model for Robust DSPN Diagnosis From Temperature Maps

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Abstract—Diabetic sensorimotor polyneuropathy (DSPN) leads to pain, diabetic foot ulceration (DFU), amputation, and death. The diagnosis of advanced DSPN to identify those at risk is key to preventing DFU and amputation. Alterations in foot pressure and temperature may help to detect DSPN and the risk of DFU. We have applied a robust machine-learning approach to identify patients with severe DSPN using standing foot temperature maps generated using temperature sensor data. A robust shallow operational neural network model DSPNet is proposed. The study utilized a labeled dataset from the University Hospital Magdeburg, Magdeburg, Germany, consisting of temperature sensor data from eight different points on the foot in seating and standing positions in patients with severe DSPN ($n = 25$) and healthy controls ($n = 18$). The proposed network achieved an F1 score of 90.3% for identifying patients with DSPN and outperformed current state-of-the-art deep-learning network methods. This is the first of its kind of research where the results confirm that temperature maps are not only effective in the detection of those at high risk of DFU but also in identifying patients with severe DSPN. Such sensors could easily be incorporated into smart insoles.

Index Terms—Deep learning, diabetic foot, diabetic sensorimotor polyneuropathy (DSPN), noninvasive diagnosis, plantar foot temperature map, self-operational neural network.



I. INTRODUCTION

ACCORDING to the International Diabetes Federation (IDF), more than 537 million people are suffering from diabetes mellitus (DM), which costs around U.S. \$966 billion in health expenditure [1]. DM leads to severe foot complications such as diabetic sensorimotor polyneuropathy (DSPN), which can lead to foot ulcers known as diabetic foot ulcer (DFU) and eventually lead to amputation and deaths [2], [3],

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[4]. Efficient, easy, and noninvasive detection of DSPN would enable timely intervention to prevent DFU. The COVID-19 pandemic made the world realize the importance of remote healthcare monitoring, avoiding the burden on the healthcare infrastructure.

It is imperative to detect diabetic foot complications—DSPN and DFU early so that advanced techniques can be used to detect these and help in remote health monitoring. Blood vessels are wrapped around the nerves in the human body and provide them with the oxygen needed to function properly. The nerves are responsible for carrying signals to and from the human brain. DM causes the accumulation of excess sugar in the blood vessels, which restricts the flow of blood. This restriction slowly damages the nerve due to reduced oxygen supply, and the damaged nerves cannot function properly [5]. The condition where the sensorimotor nerves in the extremities are damaged and thus cannot function properly is known as DSPN. In most situations, DSPN leads to altered walking patterns, frequent falls, and, in extreme cases, a lack of feeling or pain even after injuries. The latter can lead to a DFU where the patient does not notice an injury until it worsens and leads

to incurable infections because of the lack of natural healing due to the reduced blood flow. Thus, DSPN can be seen as a result of a signal propagation problem, while DFU can be seen as a result of a blood flow problem. In the literature, the proposed methods for the detection of DSPN depend on the flow of electrical signals via nerves in different parts of the foot, such as 10-g monofilament test [6], 128-Hz tuning fork [7], quantitative sensory testing [8], and the gold-standard technique of nerve conduction study (NCS) [9], [10]. DSPN leads to a loss of sensation and proprioception with a change in the walking pattern, frequent falls, and deranged tissue blood flow, especially in response to increased pressure, which leads to the breakdown of the skin and DFU. The Michigan neuropathy screening instrument (MNSI) [11], diabetic neuropathy symptom score (DNS) [12], neuropathy disability score (NDS), Toronto clinical neuropathy score (TCNS), and neuropathy impairment score (NIS) [13] enable relatively quick but a crude assessment of DSPN [14]. Quantitative sensory testing [8] and NCSs have been proposed to confirm the diagnosis of DSPN [9], [10]. More specialized techniques, such as skin biopsy with quantification of intraepidermal nerve fiber density and corneal confocal microscopy (CCM), have been proposed to enable the detection of small fiber neuropathy and early DSPN [11], [15], [16], [17]. However, all these techniques require a face-to-face consultation of patients with a health care practitioner. Besides, the DSPN diagnosis requires expensive equipment and specialized expertise in a medical facility. The most common and popular way of DSPN detection is through the nerve conduction velocity test, which is moderately painful since mild electric shock is given to the patient to determine the velocity of the signal propagation through the nerve.

The advancement in machine-learning (ML) algorithms has led to their application in various domains, especially in biomedical engineering [18], [19]. Studies using MNSI variables and ML classifiers, such as random forest algorithms, have shown that they can detect differing DSPN severity with perfect reliability [14]. Williams et al. [20] focused on automating the in vivo CCM (IVCCM) system using ML to provide a more precise, trustworthy diagnosis of DSPN. It should be noted that the equipment used for gathering the corneal images is expensive, difficult to use and require highly specialized medical personnel. Studies have shown that the pattern of plantar pressure distribution differs during gait between patients with and without DSPN [21]. Niemann et al. [22] assessed plantar temperature and found no difference between healthy controls and diabetic patients with mild DSPN. However, in our recent work, Khandakar et al. [23] found that foot thermograms can be used to distinguish between diabetic and healthy patients with a new set of features. The current study extends this approach with temperature maps generated from temperature sensor data at eight different points of the foot. In fact, the authors have been studying the use of thermograms, a noninvasive method for early diabetic foot complication detection [23], [24], [25]. The thermogram devices are expensive and need special setup. Other ways of monitoring the plantar temperature maps are

with the use of temperature mats such as Podimetric [26], [27]; socks, such as Siren or Orpyx [28], are only available in North American markets and are hardly available yet in other countries. As a result, there is a research gap of incorporating cheaper and easy-to-available devices to classify patients suffering from diabetic foot complication such as DSPN with a high accuracy performance. The temperature map can be a noninvasive and alternative tool for any corner of the globe to access.

To address the need for a robust and in-home solution, this study proposes the use of a novel network model on the temperature maps from the foot insoles embedded with temperature sensors. The proposed approach is a noninvasive and highly cost-effective approach that can be easily performed at the convenience of the patient's home. The novel and significant contributions of this study can be summarized as follows.

- 1) A complete framework has been proposed for generating a temperature map from temperature sensor data from eight critical points of the foot, which the proposed ML approach can then analyze.
- 2) An accurate and robust self-organized operational neural network (Self-ONN) [29], [30], [31], [32], [33], [34] model, DSPNet, is proposed to detect DSPN over the foot temperature maps. The proposed work is the first of its kind, suggesting using temperature maps for classifying a nerve problem, i.e., DSPN.
- 3) The state-of-the-art detection performance level is achieved with a significant gap over the competing methods.

The rest of this article is organized as follows. Section II details the proposed methodology. Section III presents the results with comparative evaluations, followed by discussions. Finally, Section IV presents the conclusions and suggests topics for future research.

II. METHODOLOGY

This study developed a novel Self-ONN model, DSPNet, to detect patients with DSPN from the temperature maps generated from the temperature sensors at the patients' foot soles. The overall methodology of the proposed framework is shown in Fig. 1. The details of the dataset used, the proposed method for generating the temperature maps, the preprocessing, and the configuration and training details of DSPNet will be covered in the rest of this section.

A. Dataset

The study used a labeled dataset of 25 diabetic patients with severe DSPN without DFU and 18 nondiabetic healthy control patients from the university clinic of Otto von-Guericke University, Magdeburg, Germany. Of the patients with DSPN, there are six females and 19 males with an age distribution of 64.8 ± 9.8 years, while the healthy participants consisted of ten females and eight males with an age distribution of 62.9 ± 7.6 years. The healthy participants did not have diabetes with no history of macroangiopathy, foot ulcerations, or prior

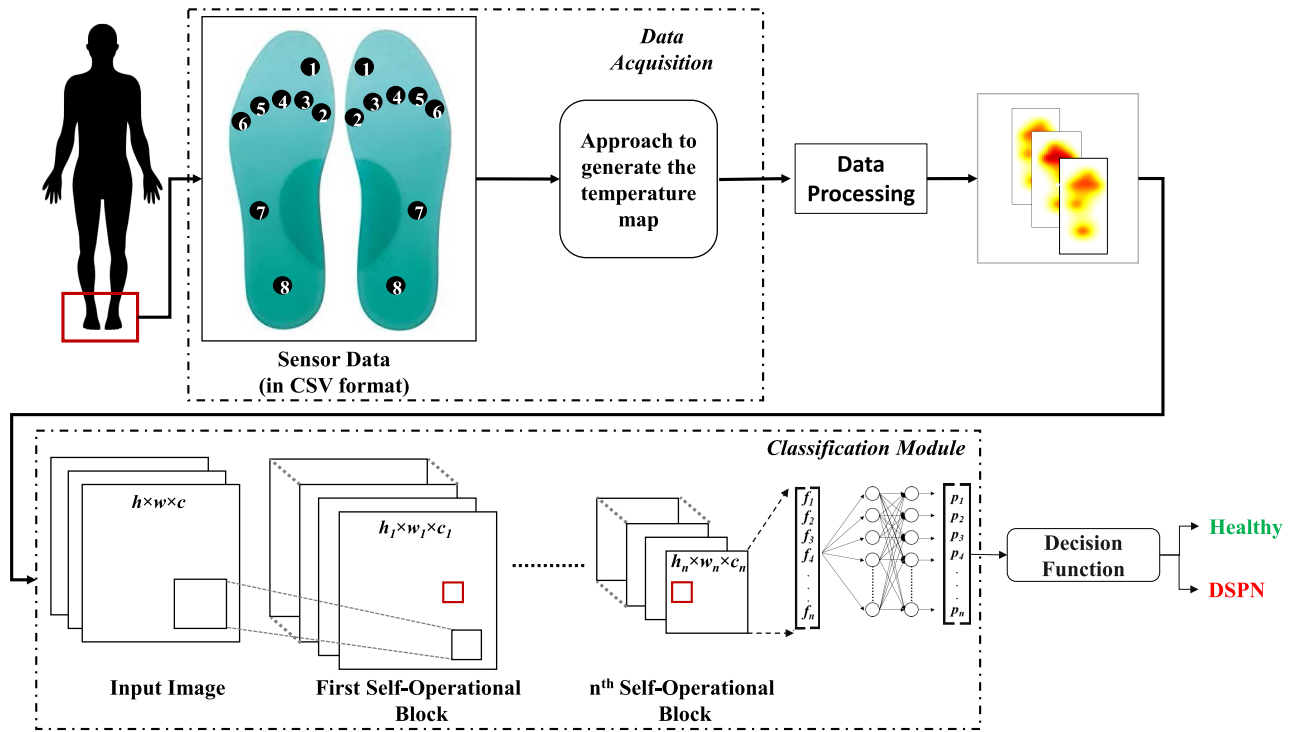


Fig. 1. Illustration of the methodology of this study.

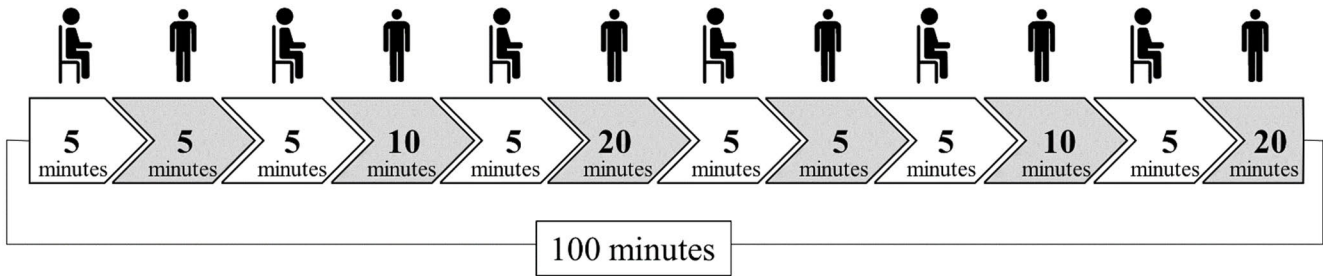


Fig. 2. Steps of the data collection session. A total of 100 min of data collection session was taken with standing and seated postures. Six 5 min of seated postures were taken as intervals between standing postures in the sequence of 5, 10, 20, 5, 10, and 20 min.

amputation. The diabetic patients had severe DSPN based on impaired vibration perception to a 128-Hz tuning fork, the absence of thermal distinction using tip-therm testing, and the absence of 10-g Semmes–Weinstein monofilament perception (threshold of $\leq 2/8$ plantar sites). A dedicated insole consisting of eight temperature sensors connected to a measurement system was used to record the temperature data for the subjects in the study [35]. The sensor placement was based on the knowledge of critical regions prone to foot deformity, refer to Fig. 1 (see the Supplementary Material) [36]. The insole comprises eight temperature sensors on the upper surface of the insole to allow close contact with the participant's foot. The sensor placement has also been investigated in some of the recent work of Mahmud et al. [37] and Khandakar et al. [38], where it has helped in generating distinguishable plantar temperature maps.

The participants were required to complete a “session” that included both seated and standing postures, while they put on the insole, for 100 min. The steps of this session were given

as follows: sit for 5 min, stand for 5 min, sit for 5 min, stand for 10 min, sit for 5 min, and stand for 20 min, as shown in Fig. 2. Sensor recordings of both feet were available for all 43 subjects (25 diabetic patients and 18 nondiabetic patients). For 15 of the 43 sessions, the raw sensor values (from the insole) were recorded every 20 s, and for the remaining sessions, it was recorded every 3 s.

B. Temperature Map Generating Approach

This section contains the data acquisition and data processing part of the methodology, as shown in Fig. 1. The authors have only considered the standing posture data as we are getting the loading condition on feet during the standing position, which was segmented using the timing information of data acquisition. The sitting position was not showing proper temperature distribution as it is not a complete loading condition and, therefore, was not considered. Some of the collected data were corrupted due to issues during data collection; thus, only data where the temperature was in the

acceptable range, i.e., [10 °C, 50 °C], are considered [39]. The thermal sensor data were extracted based on the approach, presented in Algorithm 1 (see the Supplementary Material), and extrapolated to temperature maps using a heat color map in MATLAB [40].

C. DSPNet Architecture

To overcome the limitation of the linear nature of convolutional neural networks (CNNs), operational neural networks (ONNs) can be used as presented in [41]. An ONN is a heterogeneous network that can benefit from a fixed set of nonlinear operators and has demonstrated superior results in numerous applications such as image denoising and image restoration. Self-ONNs [29], [30], [31], [32], [33], [34] are a recent variant of ONN. Instead of a fixed collection of operator libraries, a Self-ONN can learn to generate the best set of operators during the training process using its generative neurons. This ability negates the need for the prior operator search and thus can offer the utmost level of operator diversity. During the training phase, each kernel element in each generative neuron can create the best nodal operator to maximize the learning performance. The output x_k^l at the k th neuron of the l th layer of any ONN can be illustrated as follows:

$$x_k^l = b_k^l + \sum_{i=1}^{N_{l-1}} \Psi_{ki}^l (w_{ki}^l \cdot y_i^{l-1}) \quad (1)$$

where b_k^l and w_{ki}^l denote the biases and weights corresponding to that neuron and layer, respectively, y_i^{l-1} represents the previous layer's input, N_{l-1} stands for the kernel size of that layer, and Ψ_{ki}^l corresponds to the nodal operator of the neuron and layer. If Ψ_{ki}^l is the linear operator for all neurons in the network, the Self-ONN will become a conventional CNN. Instead of a fixed set of operators, the composite nodal operator Ψ can be constructed using a Taylor series approximation.

The Taylor series approximation of a function $f(x)$, near point, $x = a$ is expressed as follows:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots \quad (2)$$

Equation (2) can be used to construct the nodal operator as follows:

$$\Psi(w, y) = w_0 + w_1(y-a) + w_2(y-a)^2 + \dots + w_q(y-a)^Q \quad (3)$$

where $w_q = (f^{(q)}(a)/q!)$ is the q th parameter of the Q th order polynomial. In a Self-ONN model, tangent hyperbolic (tanh) is used as the activation function bounded at the range of $[-1, 1]$. Thus, for tanh, $a = 0$ in (3).

The Self-ONN model is evaluated with the polynomial orders, $Q = 1, 3$, and 5 , namely, Self-ONN Q1, Q3, and Q5. As mentioned earlier, with $Q = 1$ for all neurons in the network, the Self-ONN Q1 model is equivalent to a CNN. Fig. 3 shows the DSPNet configuration of the DSPNet

Algorithm 1 Temperature Map from the Eight temperature sensor data values on the foot

Input : Template with eight spots filled in, A
Output : Template with all temperature values extrapolated, A

- 1: Store the original shape of A as row (n) and column (m)
- 2: Calculate the product of row and column, $nm = n * m$;
- 3: Convert A into a column vector;
- 4: Find which positions are already known and which positions to be fulfilled and store them as *known_list* and *nan_list* respectively;
- 5: Count how many missing values and store it as *nan_count*;
 // Find row and column indices of the missing values
- 6: $[nr, nc] = \text{ind2sub}([n, m], \text{nan_list})$;
 // Create a new array *nan_list* where the 1st, 2nd and 3rd columns are unrolled, row and column index respectively
- 7: $\text{nan_list} = [\text{nan_list}, nr, nc]$;
- 8: Create *hv_list*:

$$hv_list = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 1 & 0 \\ -n & 0 & -1 \\ n & 0 & 1 \end{bmatrix};$$
- 9: Declare empty list *hv_springs*;
- 10: **for** $i \leftarrow 1$ to 4 **do**
- 11: $hvs = \text{nan_list} + \text{repmat}(hv_list(i, :), \text{nan_count}, 1)$;
- 12: $k = (\text{hvs}(:, 2) \geq 1)$
 $(\text{hvs}(:, 2) \leq n) (\text{hvs}(:, 3) \geq 1)$
 $(\text{hvs}(:, 3) \leq m)$;
- 13: $\text{append}[\text{nan_list}(k, 1), \text{hvs}(k, 1)] \text{to } hv_springs$;
- 14: $i++$;
- 15: **end for**
- 16: Remove duplicate rows in *hv_springs*;
- 17: Sort *hv_springs* along the column in ascending order;
- 18: Calculate Number of rows in *hv_springs* as *nhv*;
- 19: Create a sparse matrix, *springs*:
 $\text{springs} = (\text{repmat}([1 : nhv]', 1, 2), hv_springs, \text{repmat}([1 - 1], nhv, 1), nhv, nm)$;
 // Eliminate known values
- 20: $\text{rhs} = -\text{springs}(:, \text{known_list}) * A(\text{known_list})$;
 // Fill the missing values in template
- 21: $A(\text{nan_list}(:, 1)) = \text{springs}(:, \text{nan_list}(:, 1)) \backslash \text{rhs}$;
- 22: Reshape the column vector A to the shape (n, m)

($Q = 3$), which is the best-performing network. The network consists of five operational layers. The max-pooling layers

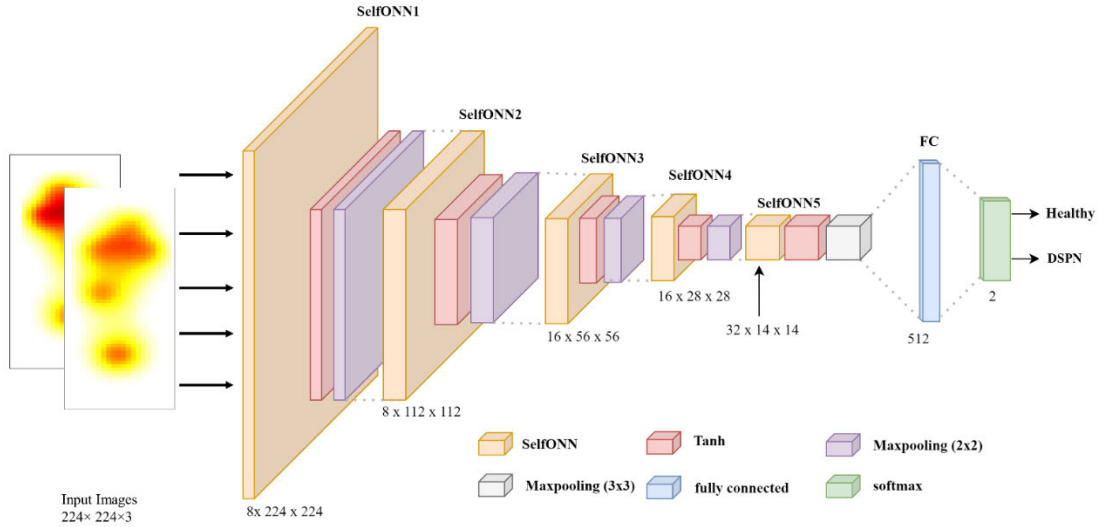


Fig. 3. Proposed DSPNet architecture.

TABLE I
DETAILS OF THE DATASET USED FOR TRAINING AND TESTING

Classes	Subjects (Images)	# of Training Subjects (Images)	# of Validation Subjects (Images)	# of Test Subjects (Images)
DSPN	20 (41,296)	16 (32,150)	2 (4,100)	2 (5,046)
Healthy	16 (15,954)	12 (12,650)	2 (1,450)	2 (1,854)

have the kernel sizes of 2 and 3, resulting in a feature embedding of 512 samples. Finally, the prediction score was computed in the classification layer with two output neurons.

D. Train–Test Fold Creation and Augmentation

In this study, tenfold patientwise cross validation was used. Therefore, 90% of patients were used for training (where 10% of training subjects' data were used for validation) and 10% for testing. In this way, the entire patient data are used for evaluation in different folds. Since the DSPN and healthy classes are not balanced, to avoid overfitting [42], three image augmentation techniques (rotation, scaling, and translation) were used. The augmentation techniques help deep learning networks to learn distinctive features on a balanced dataset. For image augmentation, the images were rotated in a clockwise and counterclockwise direction with an angle of 5°–10°. The scaling technique entails enlarging or shrinking the image's frame size, and 2.5%–10% image magnifications were used in this study. Image translation was done by 5%–10% horizontally and vertically. Table I details the number of training, validation, and test data sets used in this study. It should also be noted that some of the patients' data had to be removed as they had temperature sensor data values that were not logical, for example, temperature values of 100 °C (i.e., temperature not in the acceptable range, i.e., [10 °C, 50 °C] [39]).

E. Decision Function

For patientwise classification, the average of the prediction probability scores of different temperature map images

for each patient was used to produce the final decision. Equation (4) expresses the final classification function

$$\sigma = \frac{1}{n} \sum_{i=1}^n P_i(x) \quad (4)$$

where σ is the decision function of n number of temperature map images and $P_i(x)$ is the probability score of the i th image. The decision function σ is deciding for patientwise using the average probability score of all the patients' plantar temperature maps.

F. Visualization Techniques

The authors visually confirmed the regions of the foot from where the network makes decisions using visualization tools. The visual depiction of Self-ONN's decision-making is improved with the use of visualization tools. They increase the confidence in the Self-ONNs' outputs and the transparency of the model by explaining the interpretation behind the inference in a form that can be visualized. SmoothGrad [43], Grad-CAM [44], Grad-CAM++ [45], and Score-CAM [46] are the common visualization techniques. GradCAM is a generalization of class activation mapping [44], avoiding the accuracy against interpretability tradeoff and modifying the model architecture. GradCAM can visualize models with fully connected layers, a limitation of CAM. GradCAM visualizations for image classification clarify Self-ONN failures by demonstrating that explanations for irrational predictions may be found. GradCAM was selected for this study due to its promising outcomes in current computer vision problems.

TABLE II
COMPARISON OF THE AVERAGE PERFORMANCE METRICS FROM FIVEFOLD CROSS VALIDATION

Model	Parameters	Overall					
	Number of training parameters/ Model Size	Accuracy	Precision	Recall	F1 score	Specificity	Inference Time (ms)
MobileNetV2	2,226,434/ 53 layers (25 MB)	76.02 ± 0.35	82.25 ± 0.31	76.02 ± 0.35	78.35 ± 0.34	59.36 ± 0.40	6.06
DSPNet without Decision Function	17,130/ 5 layers (4 MB)	88.73 ± 0.26	90.80 ± 0.24	88.73 ± 0.26	89.12 ± 0.26	92.44 ± 0.22	1.48
DSPNet with Decision Function	17,130/ 5 layers (4 MB)	91.67 ± 0.90	92.98 ± 0.83	91.67 ± 0.90	93.33 ± 0.82	91.68 ± 0.90	2.45

GradCAM highlights the areas of the image that Self-ONN considers before making prediction. These visualizations help to boost confidence in the dependability of deep layer networks by validating prediction made from significant areas of the foot temperature images.

G. Performance Metrics

The comparative evaluations are performed using standard performance metrics such as accuracy (A), precision (P), recall (R), F1 score ($F1$), and specificity (S). The formulations of the performance metrics (weighted sensitivity or recall, specificity, precision, overall accuracy, and F1 score) are shown in the following equations:

$$A_{\text{class}_i} = \frac{((\text{TruePositive})_{\text{class}_i} + (\text{TrueNegative})_{\text{class}_i})}{((\text{TruePositive})_{\text{class}_i} + (\text{TrueNegative})_{\text{class}_i} + (\text{FalsePositive})_{\text{class}_i} + (\text{FalseNegative})_{\text{class}_i})} \quad (5)$$

$$P_{\text{class}_i} = \frac{(\text{TruePositive})_{\text{class}_i}}{(\text{TruePositive})_{\text{class}_i} + (\text{FalsePositive})_{\text{class}_i}} \quad (6)$$

$$R_{\text{class}_i} = \frac{(\text{TruePositive})_{\text{class}_i}}{(\text{TruePositive})_{\text{class}_i} + (\text{FalseNegative})_{\text{class}_i}} \quad (7)$$

$$F1_{\text{class}_i} = 2 \frac{P_{\text{class}_i} \times R_{\text{class}_i}}{P_{\text{class}_i} + R_{\text{class}_i}} \quad (8)$$

$$S_{\text{class}_i} = \frac{(\text{TrueNegative})_{\text{class}_i}}{(\text{TrueNegative})_{\text{class}_i} + (\text{FalsePositive})_{\text{class}_i}} \quad (9)$$

where $i = 0$ and 1 correspond to healthy and DSPN classes, respectively, true positive represents how many of the true class is correctly identified, and true negative represents how many of the true classes were incorrectly identified. False positive represents how many of the other classes were correctly identified, and false negative represents how many of the other classes were incorrectly identified.

All classification networks were implemented using the PyTorch library with Python 3.7, 64-GB RAM, and a 16-GB

NVIDIA GeForce GTX 1080 GPU using an Intel¹ Xeon¹ E5-2697v4 processor running at 2.30 GHz. The overall evaluation metrics are computed over the final confusion matrix, which is obtained by cumulating the confusion matrices of the tenfold. The best-performing model and the framework are deployed in Raspberry Pi 4 to evaluate its suitability in a low-resource system.

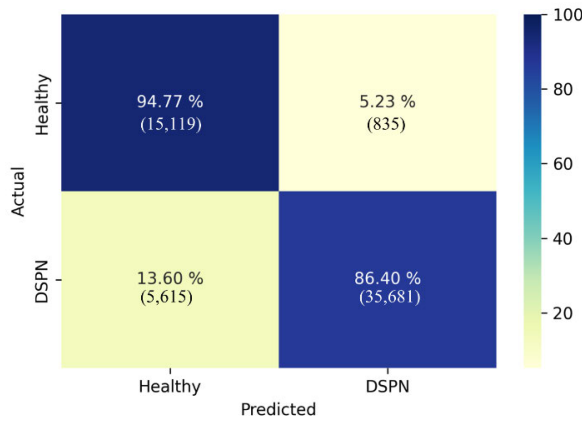
III. EXPERIMENTAL RESULTS

In this section, the performance of the trained models is shown along with the inference time of the model. An ablation study is also conducted on whether a decision function helps improve the performance of the model. Furthermore, explainable artificial intelligence (AI) methods are used to focus on the specific regions of the temperature map that contributes toward the decision of the model.

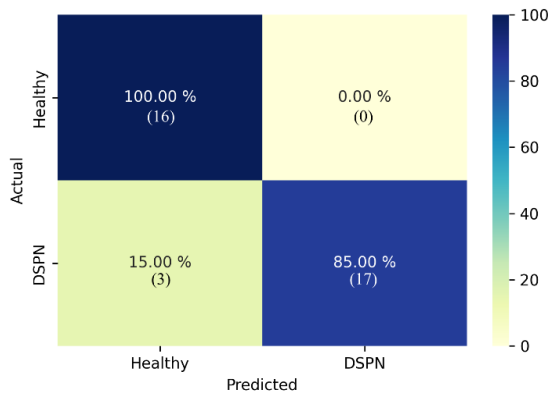
From Table II, it can be seen that the light and novel network DSPNet was outperforming the state-of-the-art MobileNetV2 [47] to differentiate healthy patients from patients with DSPN, using the generated temperature maps from the different parts of the insole (Fig. 1, see the Supplementary Material). MobileNetV2 is a light CNN network deployable on portable devices and performed well on many complicated image domain problems [24], [25].

We assessed the performance of DSPNet by comparing the effectiveness of the decision function. The decision function is best understood by considering that most of the temperature maps from a subject at different time points can be helpful to improve the performance of the network by providing more useful information rather than the single temperature map decision. A comparison was also performed between the state-of-the-art lightweight network and the proposed network in relation to the inference time, i.e., the time taken by the network to provide the results. DSPNet is a lightweight model that requires around 130 times less training parameters than

¹Registered trademark.



(a)



(b)

Fig. 4. Confusion matrix for subjectwise classification using DSPNet (a) without decision function and (b) with decision function.

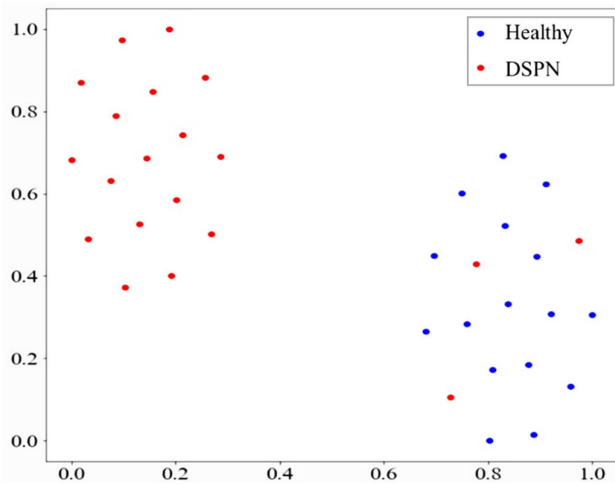


Fig. 5. t-SNE plot visualization for the healthy and DSPN classes.

the state-of-the-art MobileNetV2 model used in this study. DSPNet is around four times faster at single-sample prediction because of its smaller size, which can be inferred from the model parameters column in Table II. This means that DSPNet is more practical for use in the real world than MobileNetV2. As can be seen in Table II, the inference time increased slightly with the inclusion of the decision function, as stated earlier and represented in (4), but the performance of the model significantly improved by the usage of the decision

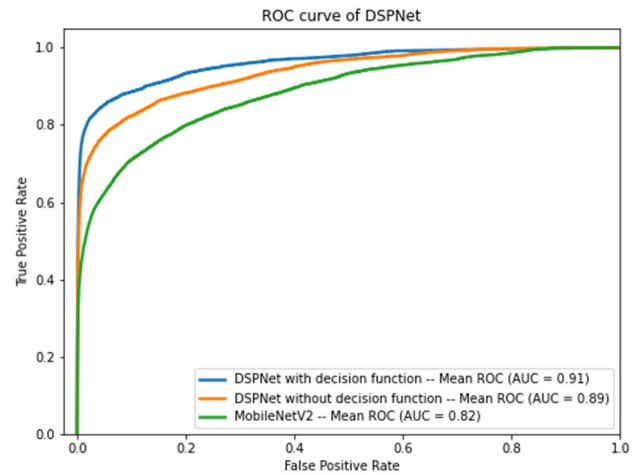


Fig. 6. ROC curve of the best-performing approaches.

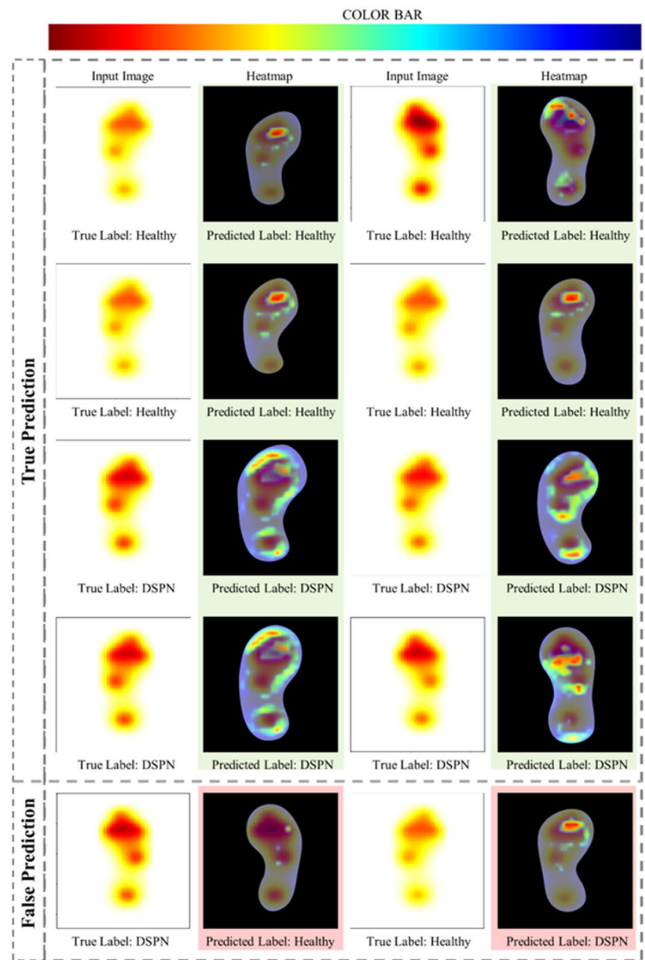


Fig. 7. GradCAM visualization of images from healthy participants and patients with DSPN.

function. The increase in inference time was due to the extra computation but was still lower than the MobileNetV2 network.

We have further confirmed the performance of the DSPNet with the help of a confusion matrix (Fig. 4). As can be seen in Fig. 4(a) and (b), the DSPNet with the decision function has improved performance. The network successfully detected all the healthy participants and only misclassified 3 DSPN patients as healthy participants, and it is worth mentioning

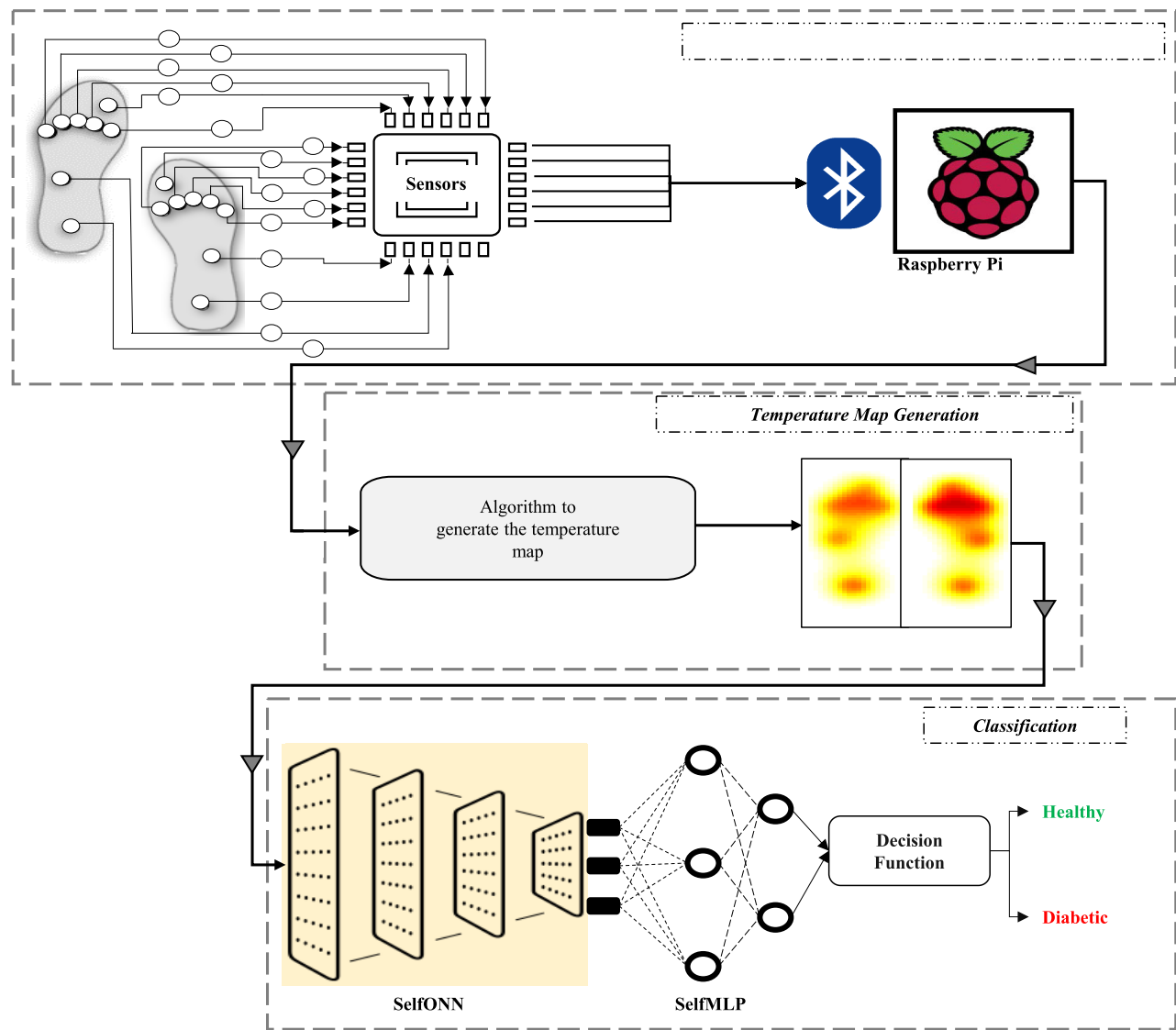


Fig. 8. Overview of the deployment phase.

here that this is the first of its kind result reported where temperature maps are used for detecting DSPN complications. The results are visibly confirmed in the t -distributed stochastic neighbor embedding (t-SNE) plot (Fig. 5) [48] using DSPNet on the generated temperature maps in healthy compared to patients with DSPN. The t-SNE plot is a statistical method to visualize high-dimensional data by giving each data point a location in a 2-D or 3-D map. This plot tries to show the clustering of different classes in reduced dimensionality, and the graphical representation shows a clear distinction between healthy controls (blue) and patients with DSPN (red).

Fig. 6 shows the DSPNet receiver operating characteristic (ROC) curve with and without decision functions. The various thresholds of the true positive rate versus false positive rate are shown in Fig. 6. This ROC curve can be used to perform a comparative analysis, showing that DSPNet with decision functions has a larger area under the curve (AUC) of 0.91 versus 0.89 for DSPNet without decision functions. An ROC is generally used to visualize the sensitivity and specificity of any model, with a higher Y-axis value representing a higher proportion of true positive instances compared to false

positive instances. As can be seen in Fig. 6, the performance of DSPNet with decision function is more sensitive and specific in DSPN and healthy case classification. DSPNet (with and without decision function) is outperforming significantly compared to a lightweight state-of-the-art counterpart network, MobileNetV2. To understand that DSPNet is learning from the important area of the image, GradCAM-based heat maps were created (Fig. 7).

DSPNet was more focused on the toe regions, the anterolateral eminence of the insole, the medial longitudinal arch, and the heel when classifying diabetes from healthy patients. The model is primarily learning to identify patients with DSPN from features given by sensors 1–6. GradCAM visualization of most of the correctly classified DSPN samples showed dominated heatmaps around sensors 1–6 and minor heatmap presence around sensors 7 and 8. If the model could not extract features from sensors 1 to 6, it incorrectly identified the case as healthy (Fig. 7). In other words, the sugar accumulation on the blood vessels leading to nerve problems (DSPN) also results in abnormal temperature distribution, which is easily identified by DSPNet, as can be seen from the heatmap used

by DSPNet in classifying DSPN and healthy patients. Because DSPNet learned from features other than those associated with diabetes, it categorized healthy cases with a high degree of accuracy and highly focused on sensors 6–8 (Fig. 7). If the model incorrectly classified healthy cases as having diabetes, then the heatmap shows sensors 1–6 regions rather than the sensor 6–8 regions, markers for healthy case identification. By using GradCAM visualization, it is clear that the model distinguishes between those with DSPN and healthy controls, using completely different characteristics, thus increasing user confidence in the model.

Furthermore, an affordable Linux-based single-board computer, the Raspberry Pi (RPI) 4B is used to deploy the model to see the deploy ability of the proposed system in low-configuration hardware. We have deployed the framework, which will allow the temperature sensor data to be sent over Bluetooth to the RPI and generate temperature maps that are then fed to the DSPNet and decision function to provide the results. The algorithm, DSPNet, and the decision function are deployed in Raspberry Pi, where our AI-based application analyzes the data to determine whether the subject is healthy or a DSPN patient. Fig. 8 shows the overview of the deployment phase.

In the prototype system, the temperature sensor values from the eight critical temperature points are sent to the Raspberry Pi. We have already proposed how a smart foot sole can be created to allow data to be wirelessly transmitted to a processing unit [49]. Therefore, the details of the sensor system are not discussed in this work. The algorithm to convert the sensor data to a temperature map is deployed in Raspberry Pi. It uses multiple temperature maps acquired over 30 s to be applied to the trained network (with a decision function) to generate the prediction, and the prediction is displayed on the screen. In summary, the application can promptly assess the patients using the temperature from eight critical points on foot. This can help in reducing the need for face-to-face consultations required to undertake a monofilament or neurological exam, thus reducing the burden on the healthcare system.

IV. CONCLUSION

DSPN is among the alarming complications of diabetes that can lead to DFU, amputation of the foot, and eventually death if not detected and treated early. There are many techniques to detect DSPN, but they require expensive hospital or clinical settings, face-to-face consultation, and expert examination that are not convenient for patients. This article proposes a technique based on ML combined with the temperature data acquired from the foot of a diabetic patient that can be easily collected from smart insoles with embedded temperature sensors. We propose a novel ML network, “DSPNet,” which, when used in conjunction with a decision function, can differentiate DSPN patients from healthy subjects with very high accuracy. The GradCAM visualization backs up these assertions. Furthermore, the proposed algorithm is deployed in RPi4, which is a low-cost, low-resource single-board computer that can be used by the patients at home and hospital by both patients and medical doctors to conveniently identify DSPN. In future, the system can be deployed to the cloud to remove

the need for a Raspberry Pi. This work is the first study to use temperature maps to detect severe DSPN that does not require face-to-face interaction to detect DSPN, avoiding both the expensive equipment and healthcare settings as well as the expertise-needed diagnosis approaches. The current work, along with the authors’ previous work [23], [24], [25], can be used to investigate the possibility of high-resolution thermogram generation using only the data from limited sensor points and the possibility of generating dynamic temperature maps (temperature maps during walking) and using them to detect DSPN severity.

DATA AVAILABILITY STATEMENT

The data used in the study are obtained from the University Hospital Magdeburg, Magdeburg, Germany [22].

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